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1.0 Introduction

Data science relies heavily on accurate data in order to generate effective and helpful insights. When data is messy in structure and lacks of accuracy, it will weaken the analysis and reduce the reliability of results. This is where the data preprocessing plays an important role. It is a process that transforms unstructured data into a standardized and complete form that is ready for analysis. The four key steps in preprocessing include data cleaning, integration, transformation and reduction. It helps to correct errors, minimize inconsistencies and handle missing values to ensure data integrity (Konda, 2022).

Data quality refers to how well the characteristics of a dataset fit the purpose it is being used for (Zhang et al., 2021). Poor data quality will distort analytical results, waste resources and cause financial losses in the business context. Studies show a strong relationship between data quality and decision-making efficiency, low data quality results in ineffective and unreliable decisions that negatively affect the organizational performance (Alabduljabbar, 2024). Therefore, applying data preprocessing and quality management practices is important to reduce mistakes and produce more valid and dependable analysis results.

2.0 Methods by Data Type

2.1 Structured/Tabular Data

Structured data is stored in a well-defined and organized format, such as in tables, spreadsheets or databases, the data can be numeric or textual. There are predefined fields to categorize each record to ensure consistency.

Handling missing values is important in preprocessing as they can significantly affect model's results (Koukaras & Tjortjis, 2025). Deletion is a direct method by solely deleting records that contain missing data. However, it is not recommended as crucial information might be deleted. Imputation is a better approach by replacing missing data with a substitute entry through estimation based on data context. Simple imputation may use mean, median or mode to predict the missing record (Afkanpour et al., 2024).

For textual data, encoding is used to convert text into numeric data as some analyses only support numeric data types. One-hot encoding is used to transform nominal categorical data by creating a new column for each category. If a record belongs to that category, it will be denoted as 1, else, it will be denoted as 0 (Bala & Behal, 2024). The data is now in binary format instead of textual.

Normalization is a type of scaling that makes the values fit within a specific range. Without it, features with larger ranges could dominate the model, leading to biased results. Thus, it helps ensure each feature contributes fairly. Min-max normalization scales values by making the smallest becomes 0, the largest becomes 1, and all other values fall proportionally in between. Whereas z-score normalization standardizes values based on mean and standard deviation calculations (Cetin & Yildiz, 2021).

2.2 Unstructured Textual Data

Unstructured textual data are text that is not stored in any fixed format or table, like emails, reviews or social media comments. Data is raw and inconsistent, making it difficult for machines to analyse its meaning.

One of the preprocessing steps for unstructured textual data is changing all the texts into lowercase. Machines treat lowercase and uppercase as different characters although they are the same word with the same spelling (Tabassum & Patil, 2020). By lowercasing all the texts, machines are able to detect and read which text is the same and treat them consistently.

Unstructured textual data appear as long strings, making them hard for machines to process. Tokenization is used to split a sentence into individual words, characters and punctuations, called as tokens, based on splitting criteria such as whitespace or punctuation (Tabassum & Patil, 2020). This breakdown makes it easier to remove unwanted words. Some tokenizers use dictionaries to split text for languages like Chinese and Japanese, where spaces are not used to separate words (Razzaque & Basak, 2023).

Lemmatization is the process of converting a word into its base form. For example, the word “running” will be converted into “run” (Aliero et al., 2023). In natural language, a word can appear in many different forms depending on the grammar. Lemmatization helps simplify text to reduce redundancy and inconsistency, allowing the model to run faster and process data more efficiently.

2.3 Image Data

Image datasets usually contain inconsistent pixel intensity values and variations in light that will affect the model performance. Normalization such as min-max scaling and Z-score normalization ensures that pixel values are comparable by scaling pixel values to a fixed range or centered at zero and based on unit variance.

Additionally, poor lighting, weather, sensor temperature, dust, and network corruption in images can reduce reliability. The most common types are Gaussian noise that is random brightness variations, salt-and-pepper noise comprising random black and white pixels, Poisson noise that is signal-dependent, and speckle noise that may arise in radar or medical images. Denoising helps to remove unwanted noise while remaining important details such as texture and brightness. Such example techniques include mean filters, which smoothen pixels by averaging neighbours; median filters replace each pixel with the median of surrounding values, while transform domain approaches such as Wavelet Transform use fixed formulas whereas ICA adapts how it applies the formula based on the structure in each image (Swamy & Kulkarni, 2020).

Limited labelled samples in image datasets can lead to bias and overfitting (Goceri, 2023). Image augmentation helps artificially increases the dataset size by flipping, cropping, scaling, adjusting colours, randomly erasing, and mixing images. More advanced techniques include AutoAugment, which automatically seeks out the best transformations, Feature Augmentation, which modifies learned features to create new data, and Generative Adversarial Networks (GANs), which generate realistic synthetic images to further diversify training datasets (Yang et al., 2022).

2.4 Audio Data

Audio data has a lot of variations due to noise interference such as background sounds like fan, variations in recording length, inconsistent formats or encoding such as MP3 and WAV and acoustic variations because of speakers, environment or recording devices. Segmentation of audio improves quality by breaking continuous signal into smaller segments that can be analysed separately. Firstly, it filters unwanted signal, reduces background noise, and performs quality enhancement. This audio is then converted into a digital format suitable for analysis. Fast Fourier Transform (FFT) is then used for extracting features of frequency. The signal is divided into small segments, and deep learning model like Recurrent Neural Networks (RNN) is used to identify the type of sounds. Finally, segment boundaries are detected to assign each segment a type of sound and create labelled streams (Aggarwal et al., 2022).

Artificial noise in the communication channel and natural noise in the environment can reduce the audio quality, thereby distort the signal and affect the speech that will be transmitted and heard by humans. Noise suppression enhances audio quality by reducing irrelevant background sounds to make

speech clear and intelligible. For neural network processing, the audio is converted into a 2D power spectrum to show how the frequency changes over time. A Variational Autoencoder (VAE) with convolutional filters is trained to distinguish between speech (useful signal) and noise (unwanted sound), learns patterns to filter noise in new audio and then reconstructs it into a cleaned waveform (Nogales et al., 2024).

2.5 Video Data

Video data is a sequence of images captured over time where the frames are usually at 25-30 fps or higher. This creates high-dimensional, temporally and spatially rich streams (Kar et al., 2024). Video complexity is explained by its volume, and spatio-temporal variation. Preprocessing is necessary to improve the data-quality and efficiency because of large volume and redundancy.

First preprocessing method is frame sampling where a subset of frames is selected from a complete video to decrease temporal redundancy and computational loads while keeping the important content (Zhanga and Zhang, 2024). The first step is extracting frames from the video and adjusting the frame rate. Then, CLIP or Vision Transformers (ViT) models are used to extract visual features where it will then be compared using measures like cosine distance so that significant changes are identified in scene or content. Repetitive frames are removed while significant ones with semantic differences are kept as keyframes. Shorter sequences that maintain main visual information for later tasks will be the results.

The next method is temporal segmentation, also known as shot-boundary detection, where it breaks a continuous video into meaningful parts such as shots or scenes (Kar et al., 2024). It starts with using histograms or edge change ratios (ECR) to extract features from consecutive frames. The similarity between frames are calculated. Hence, when the difference exceeds the threshold then the frame is marked as boundary. Detected boundaries separate the video into coherent shots or scenes that allow for structured organization and easier analysis.

By eliminating redundant information, improving temporal organization and ensuring only relevant information is kept, these methods improve data quality.

2.6 Time Series/Sensor/IoT

Each data point in a time series is associated with a specific moment or time interval (Ciburror and Iannace, 2021). Consecutive observations in these data have dependencies with past values affecting the future ones. Analysts can predict trends by understanding these patterns as it helps with forecasting despite its randomness and uncertainty. This task is difficult due to high variability and unpredictability.

One of the preprocessing methods for time series is resampling where it resolves irregular or imbalanced time intervals in IoT and sensor data (Christopher et al., 2021). The first step is organizing raw sensor data according to timestamps. Interpolation methods like linear, spline or polynomial interpolation will be used to estimate missing values when there are data gaps or uneven sampling rates. Oversampling methods like Synthetic Minority Oversampling Technique (SMOTE) or undersampling to balance frequency of rare events is used when there are imbalance datasets. Once resampled, the data is aligned to a uniform time grid where consistency is ensured across signals for accurate trend analysis and training.

Another method is sensor drift correction where it focuses on gradual changes in IoT sensor readings caused by environmental factors or calibration loss (Hossain et al., 2024). The first stage is monitoring sensor output to read the shifts from baseline readings. Variational autoencoder (VAE) model is trained to reconstruct normal signal patterns. Possible drift is shown by differences between the reconstructed

and real signals. Recalibration algorithms adjust sensor outputs back to expected levels when drift is detected to restore accuracy.

To ensure robust forecasting, anomaly detection and data-driven decision-making, reliability, consistency and accuracy of time series and IoT data needs to be improved using these preprocessing methods.

2.7 Graph/Network Data

Graphs are composed of nodes and edges, which are used to represent entities, and the connections between the nodes respectively (Asara Senaratne et al., 2022). Data that are graph-structured tend to face quality issues such as data imbalance, whereby training data are disproportionately allocated across classes, resulting in the bias toward the majority class and failure to identify minority nodes (Tian et al., 2025). To resolve this issue, the resampling method was adopted to generate a subset sample of nodes and edges from a large graph whilst still attaining its properties (S. Dizaji et al., 2024). An example of this technique includes GraphSmote which uses the Synthetic Minority Oversampling Technique (SMOTE) to create generated data samples from the existing minority class and is often used for fake account detections in social networks (Tian et al., 2025).

Another quality problem is missing or noisy data as graphs are particularly susceptible to any feature abnormalities and disturbances (Wang et al., 2024). Song et al. (2021) explained that graphs in real world context are incomplete with missing attributes of nodes and edges. To handle missing data for graph structures, You et al. (n.d.) noted that there are two approaches. The first is feature imputation, whereby values are predicted using existing observed data. Meanwhile, another method is label prediction in which the model continues to make predictions directly from the incomplete data. Embedding-based methods are also used to handle and predict missing data based on existing data. However, errors can also occur from these methods due to the assumption that input data are correct without considering other quality issues (Xue & Zou, 2022).

3.0 Process Diagrams

3.1 Structured/Tabular Data

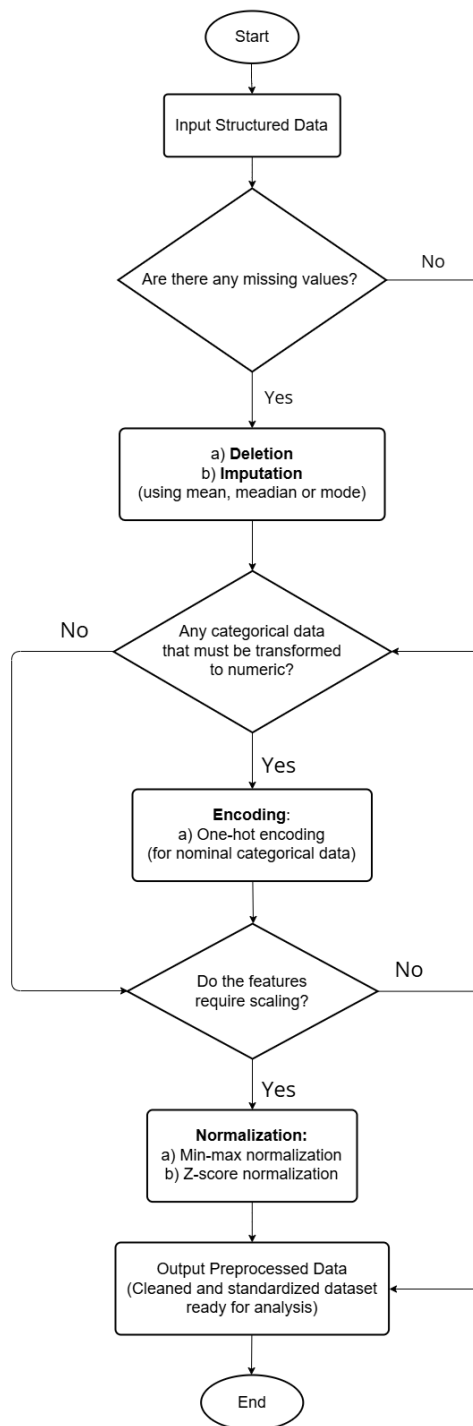


Figure 1: Structured/Tabular Data

Figure 1 illustrates the preprocessing of structured data. First, data cleaning is performed to handle missing values through deletion or imputation. Categorical features are converted to numeric format using one-hot encoding and different features scales are adjusted using normalization technique to ensure consistency. The dataset is then ready for analysis, with improved completeness, consistency and reliability of the data.

3.2 Unstructured Textual Data

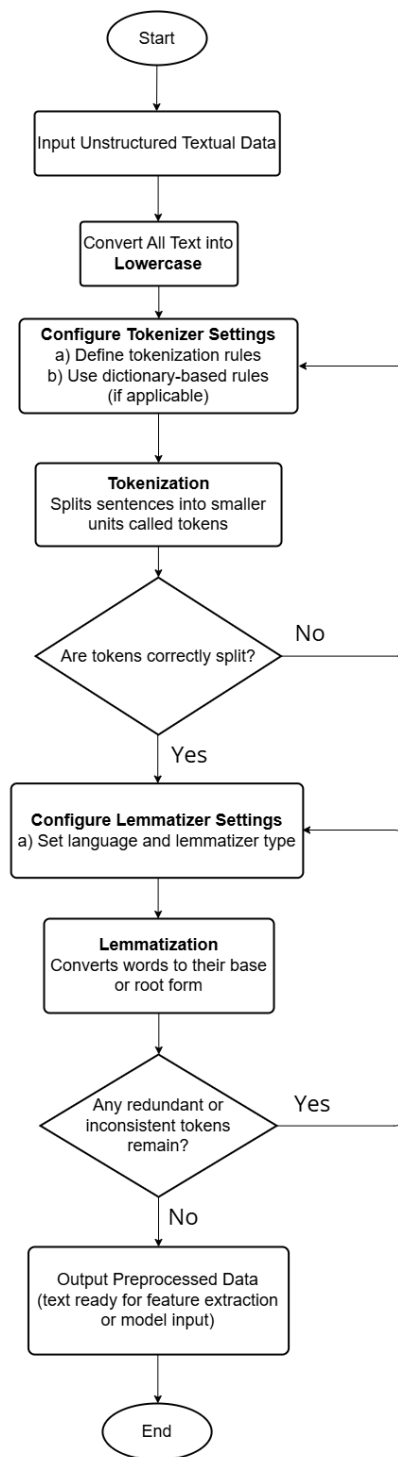


Figure 2: Unstructured Textual Data

Figure 2 illustrates the preprocessing of unstructured textual data. First, all text data is converted into lowercase. Then, tokenizer settings are configured using rules or dictionaries and adjusted if tokenization is inaccurate. Lemmatization comes next to convert words into their base forms. The dataset is then ready, with standardized words, precise tokenization and reduced redundancy.

3.3 Image Data

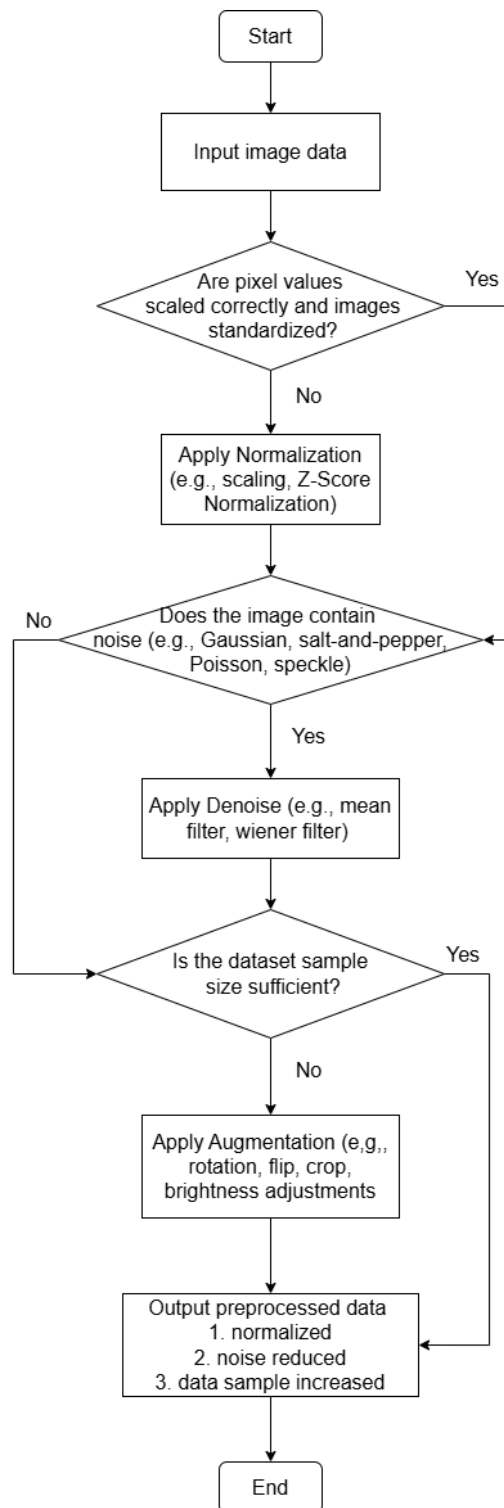


Figure 3: Image Data

Figure 3 shows image preprocessing flowchart, which combines normalization, denoising and augmentation methods. The process begins with verification of pixel values and standardization. After identifying noise types and dataset size, normalization is used to standardize the data, denoising is used to remove noise, and augmentation is used to increase dataset diversity.

3.4 Audio Data

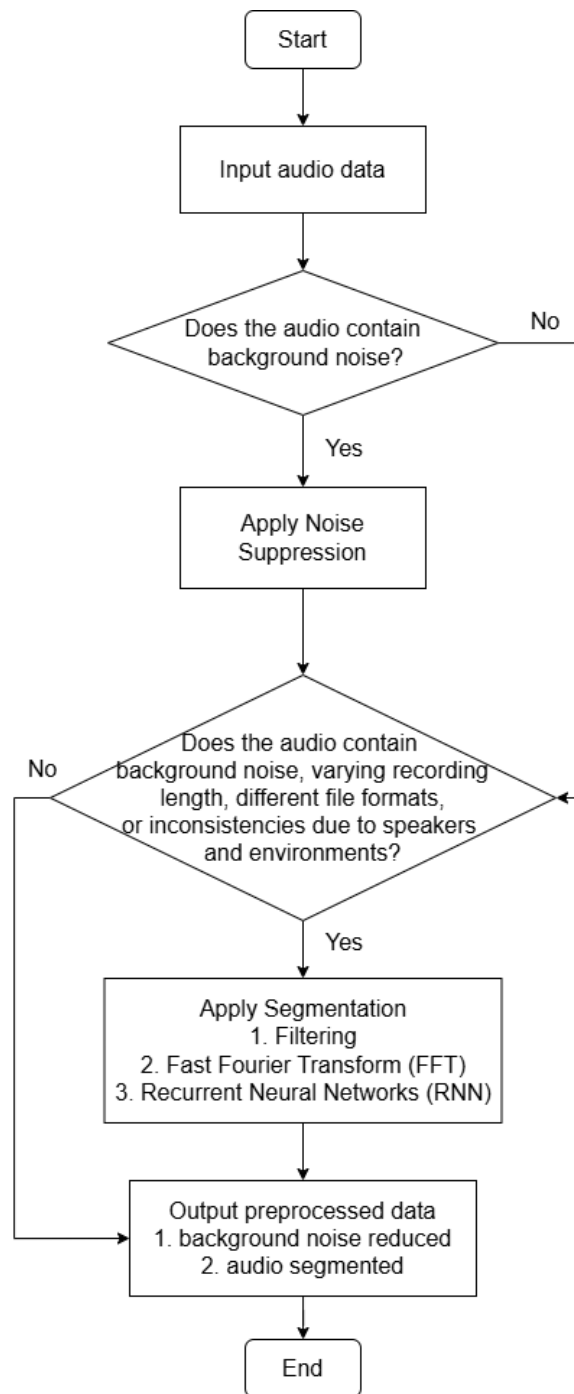


Figure 4: Audio Data

Figure 4 illustrates the audio preprocessing process, which combines noise suppression and segmentation methods. After checking for quality issues, noise suppression is performed to remove background noise. The segmentation steps include filtering, extracting features using FFT and classifying each audio segment using RNN to generate a labeled audio stream that can be used for analysis.

3.5 Video Data

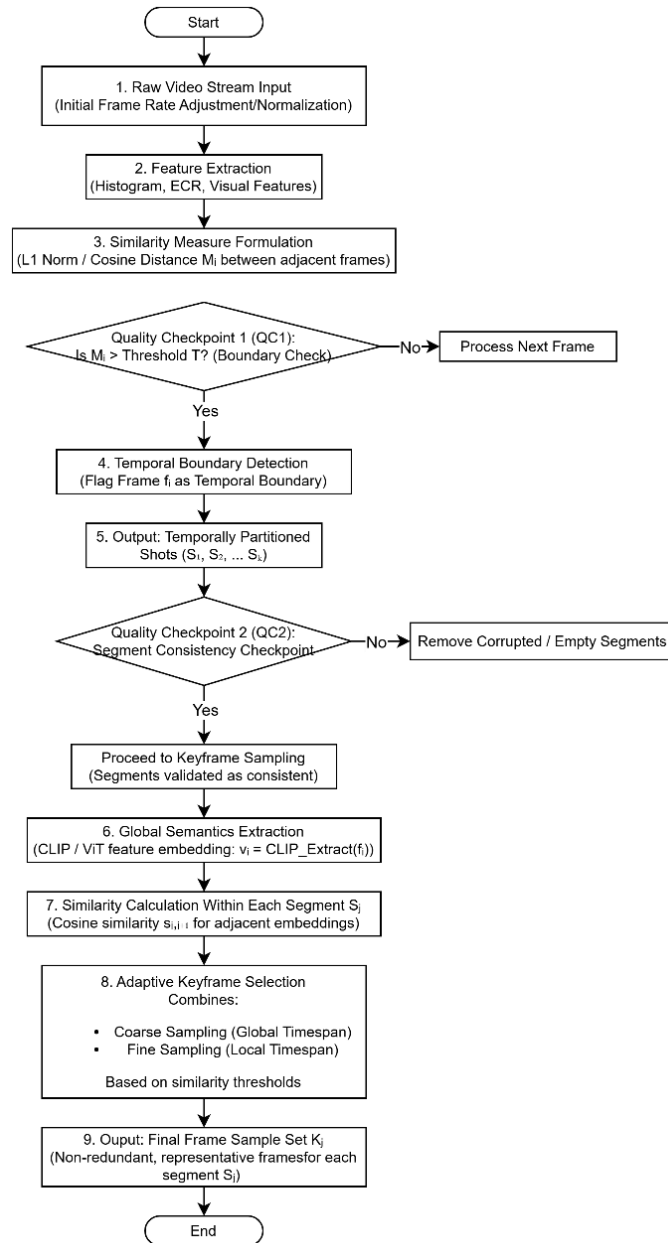


Figure 5: Video Data

Figure 5 illustrates a video-data preprocessing workflow combining temporal segmentation and frame sampling. Feature extraction and similarity measurement identify shot boundaries through quality checkpoints (QC1, QC2) ensuring consistency. CLIP/ViT-based semantic extraction and adaptive keyframe selection reduce redundancy, enhance video-data quality, preserving key content, and improving efficiency for analysis or retrieval.

3.6 Time Series/Sensor/IoT

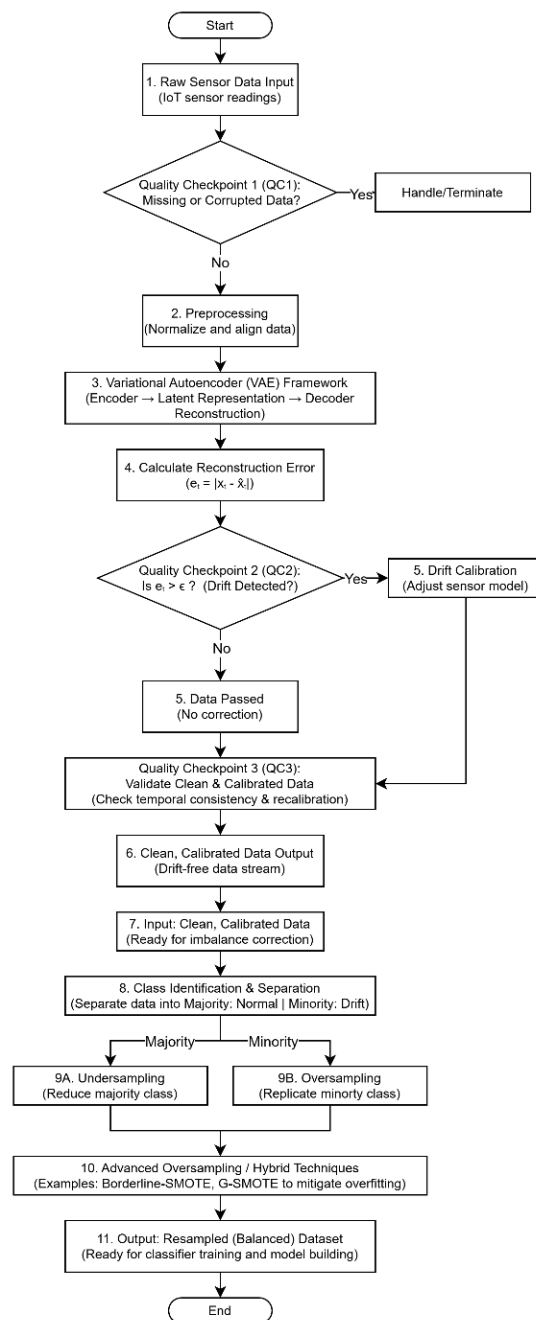


Figure 6: Time Series/Sensor/IoT

Figure 6 shows the integrated preprocessing pipeline for time-series and IoT data combining sensor drift correction and data imbalance handling. Missing or corrupted readings are detected (QC1), normalized, and processed using VAE for drift detection (QC2 till QC3). Data balancing uses undersampling, oversampling and hybrid SMOTE methods to produce clean, reliable and representative datasets.

3.7 Graph/Network Data

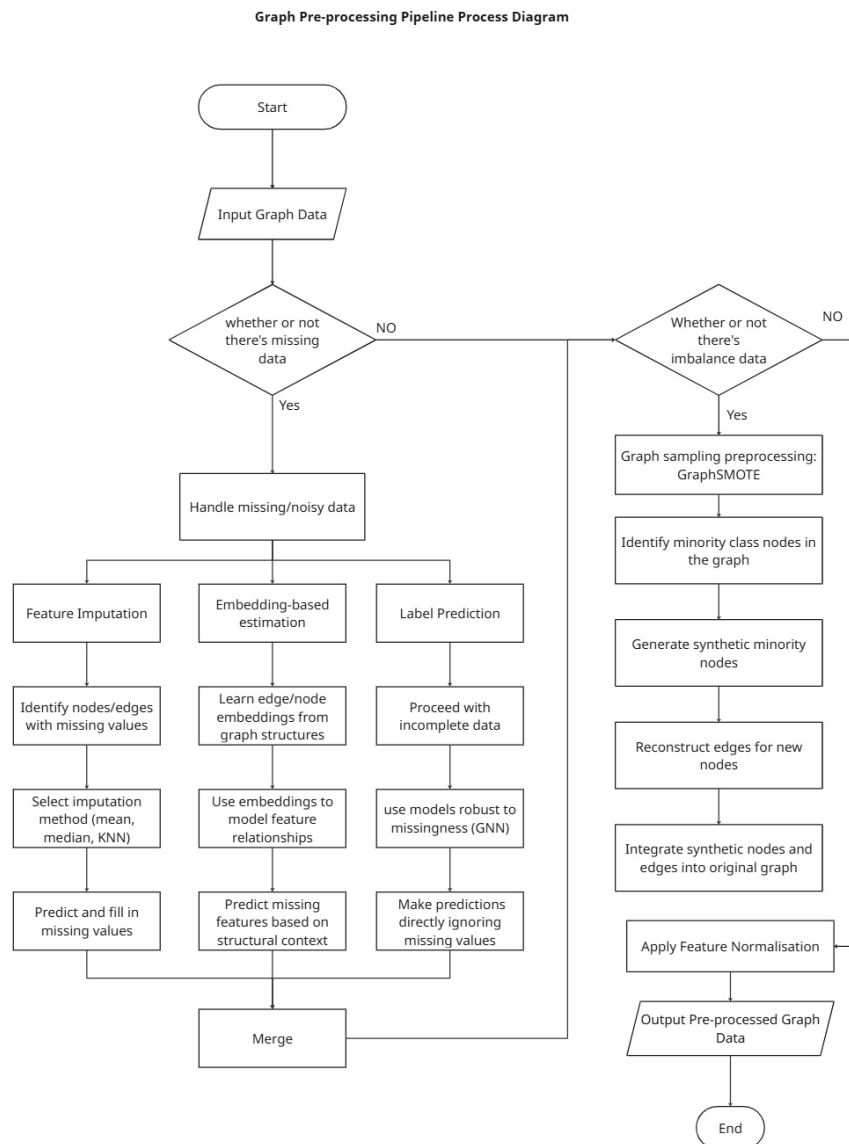


Figure 7: Graph/Network Data

Figure 7 above begins by first inputting the graph data, followed by checking for missing data attributes. To handle missing data, techniques including feature imputation, embedding based estimation and label prediction are executed. Next, the graph data is checked for imbalance data classes which can be handled with the graph sampling method, GraphSMOTE. Feature normalisation is then executed to ensure model stability.

4.0 Comparative Analysis Table

Table 1

Comparative summary of preprocessing methods across data types.

Data Type	Pre-processing Method	Data Quality Issue	Advantages	Limitations	Example Application
Structured/Tabular Data	Deletion	Missing values	Very simple and direct; no modelling required; removes missing records quickly	Can cause bias and Loss of information.	For panel health-behaviour data with <5% missing values, deletion produced results similar to imputation methods (Kim & Kim, 2020).
	Imputation	Missing values	Retains more of the dataset compared to deletion	Poor imputation may distort distributions or introduce bias	Healthcare analytics: missing discharge data imputed before predictive modelling (Wang et al., 2022)
	One-hot Encoding	Non-numeric categorical variables	Allows categorical features in models requiring numeric input	High cardinality leads to too many columns, increasing training time (Koukaras & Tjortjis, 2025)	Converted categorical features in heart failure dataset into binary variables for analysis (Prasetyo et al., 2023).
	Normalisation	Feature scale imbalance	Ensures features are on the same scale, improving fairness across attributes	Not useful for scale-invariant models (e.g. tree-based); increases complexity (Koukaras & Tjortjis, 2025)	Applied in K-means clustering to scale features similarly, preventing any feature from dominating distance calculations (Wongoutong, 2024).
Unstructured Textual Data	Lowercasing Text	Inconsistent case sensitivity	Simplifies vocabulary; easy to implement	Some uppercase carry meaning like named-entities, convert to lowercase may lose that information	Lowercasing reduced vocabulary size and improved classifier performance in Vietnamese social media sentiment analysis (Duong & Nguyen-Thi, 2021)
	Tokenisation	Unstructured long text	Facilitates further steps like stop-word removal and lemmatization	Language-dependent rules may cause tokenization errors (Razzaqe & Basak, 2023)	Splits user inputs into tokens, allowing the chatbot to identify intents and respond accurately (Naresh et al., 2023).
	Lemmatization	Redundant morphological variations	Reduces the vocabulary size as same word with different forms are reduced to the same base form	Requires language-specific rules; computationally heavier than stemming (Aliero et al., 2023)	Used in search systems to normalize words, ensuring different forms match and improving retrieval accuracy (Boban et al., 2020).
Image	Normalisation	Inconsistent pixel intensity and scale	Reduces bias across datasets; improves accuracy and convergence stability (Singthongchai & Wangkhamhan, 2025)	Not necessary for tree-based models due to scale-invariant; may increase complexity without benefit (Koukaras & Tjortjis, 2025).	Chest X-ray classification for consistent brightness (Singthongchai & Wangkhamhan, 2025)
	Denoising	Noise from lighting, dust, temperature, or transmission errors	Preserves important details (Swamy & Kulkarni, 2020); enhances algorithm performance (Dong & Kotenko, 2025); reduces post-processing errors	May cause artifacts or blurring, reducing visual fidelity (Swamy & Kulkarni, 2020)	Improves accuracy in MRI and CT scans, reduces noise in satellite imaging, enhances faint telescope signals, and improves low-light visual quality in photography and television (Swamy & Kulkarni, 2020).
	Augmentation	Limited labeled samples; data imbalance	Increases data diversity; improves performance and robustness (Yang et al., 2022)	May produce unrealistic or limited variations (Rasool et al., 2025)	Autonomous driving: create extra training samples for traffic cones and warning triangles. (Li et al., 2022). Medical imaging like X-rays (Makhlouf et al., 2023).
Audio	Segmentation	Long or continuous audio streams	Improves model training; easier handling of long recordings	Difficult to determine clear segment boundaries; overlapping sounds (Mariotte et al., 2024)	Music/speech detection: separate radio speech and music segments (Venkatesh et al., 2021). Healthcare: classify heart sounds as normal or abnormal (Aggarwal et al., 2022).
	Noise Suppression	Background or environmental noise	Improves clarity and intelligibility in noisy environments (Nogales et al., 2024)	Struggles under complex transitions (Ex: fade-ins, overlapping effects), leading to false or missed boundaries.	Real time digital audio communication such as streaming or voice transmission (Nogales et al., 2024). --- this one no need for more elaboration la Air traffic control: Clearer pilot-controller communications (Wu et al., 2024).
Video	Frame Sampling	Redundant frames and excessive video size	Reduces temporal redundancy and computational load while preserving key content, minimizing latency and storage needs (Zhang & Zhang, 2024)	May overlook fine-grained temporal cues	Applied in film editing, sports highlight generation and content indexing to efficiently detect scene changes.
	Temporal Segmentation	Unstructured video sequences	Organizes video into coherent segments; improves analysis consistency (Kar et al., 2024)	Sensitive to illumination or motion transitions	Scene boundary detection and video summarization
Time-series, sensors, IoT	Resampling	Imbalanced or inconsistent time intervals	Enhances data quality by addressing imbalance and inconsistency, improving model reliability (Christopher et al., 2021)	Man introduce synthetic noise or distort temporal dependencies, reding the accuracy of pattern recognition and anomaly detection.	Used in predictive maintenance and smart healthcare to balance irregular sensor readings for reliable modelling.
	Sensor Drift Correction (VAE-based)	Sensor aging and calibration loss	Compensates for gradual deviations in sensor readings, improving long-term accuracy and stability of IoT data (Hossain et al., 2024).	Requires high computational resources and accurate baseline models, limiting real-time deployment in large scale IoT systems	Applied in air-quality monitoring and industrial automation to maintain sensor precision and operational safety.

Graph/ Network	Graph Sampling (GraphSMOTE)	Class Imbalance	Balances class distribution while maintaining graph structure (Tian et al., 2025)	May introduce synthetic noise or distort true relationships between nodes (Wang et al., 2024)	Used in social networks, fraud detection, and citation networks to correct class imbalance and improve GNN performance (Tian et al., 2025).
	Feature Imputation	Missing Node/edge attributes	Improves completeness and robustness (You et al., n.d.)	May introduce synthetic noise or distort true relationships between nodes; computationally expensive for large graphs (Wang et al., 2024).	Applied in sensor networks, biological interaction graphs, and traffic prediction where missing attributes are common (You et al., n.d.)
	Label Prediction	Missing or partial labels	Enables training on incomplete data (You et al., n.d.)	Low accuracy when missingness is extensive; models may overfit to incomplete patterns (Song et al., 2021)	semi-supervised learning for social graphs, citation networks, and recommendation systems (You et al., n.d.).
	Embedding Methods	Noisy or incomplete graph data	Efficient similarity estimation and missing data prediction (Xue & Zou, 2022)	Assumes input data are correct; embeddings can capture and amplify noise or bias (Xue & Zou, 2022; Wang et al., 2024).	Applied in knowledge graphs, link prediction, and node classification tasks (Xue & Zou, 2022).

Table 1: Comparative Analysis Table

5.0 Discussion of Emerging Trends

In recent years, automated and adaptive frameworks that minimize human intervention and enhance scalability have started to replace manual, rule-based pipelines in data preprocessing. For example, Mumuni and Mumuni (2024) reviewed techniques that automate cleaning, imputation, encoding, augmentation and feature construction. Similarly, by allowing models to acquire representations from unlabelled or weakly-labelled data, the field of self-supervised learning (SSL) emerges as a potent technique for improving data quality, thereby tackling noise, missingness, and feature sparsity (Zheng et al., 2025). Furthermore, the expanding field of AutoML , automated machine-learning, is moving beyond model-selection into pipeline automation, which allows optimization algorithms to dynamically modify preprocessing steps like feature scaling and outlier detection instead of hand-coding them (Baratchi et al., 2024). Advantages of these advancements are faster deployment, reduced bias from human error and increased reproducibility. However, there are still issues to be resolved such as making sure automated conclusions are understandable, maintaining fairness across diverse data types and managing unintended artifacts from synthetic data. Future research should focus on monitoring automated preparation in production settings, ethics, and transparency.

6.0 Critical Evaluation and Integration

Data preprocessing is important across various data types, whether textual, audio, images, video or graphs. Although the methods are different across the various data types, these methods share similar purposes. For instance, noise reduction or data cleaning in images, audio, and video eliminate unwanted details to make analysis more accurate and reliable. Normalization in structured data and images ensures that no feature or value contribute unfairly to models, which will lead to bias. Moreover, segmentation or tokenization is used in textual, audio, and video data as a method of breaking data into units for easier processing to produce more accurate analysis. Furthermore, handling missing or imbalanced data in structured and graph data aims to improve completeness and reduce bias. The objective of data preprocessing is to have accurate, consistent, complete, and reliable data. Preprocessing cleans the noise, handles missing or imbalanced data, normalizes or standardizes values, and structures it into units for analysis. The technique changes raw and inconsistent data into usable formats that improve data quality, reduce redundancies, save time and cost, and thus allow for more accurate and reliable analyses that can be used in machine learning and research studies.

7.0 Conclusion

In conclusion, this paper aims to evaluate the different pre-processing techniques to address the data-quality issue of a variety of data types including structured, textual, image, audio, video, time-series, and graphs. Effective data pre-processing is an important step in data mining and data science as it not only improves model performance and accuracy but also can reduce model complexity and performance bias. As technology and the need for data continues to grow, emerging trends such as

automated adaptive frameworks and AutoML are expanding to meet these needs. This significantly contributes to improved productivity and opportunities for scalability across industries. Examples of practical applications include using the denoising method in image data types to reduce atmospheric and noises for satellite or GIS imaging as well as applying the frame sampling technique in video data types for film editing to detect for changes in the video scenes. Ultimately, while different pre-processing techniques have their advantages and limitations, applying the appropriate methods can help to ensure optimal model performance analysis.

(2998 words)

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